Chaoyi Wang

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EDUCATION

Ph.D. Candidate, Communication and Information Systems — 2024 – Present
University of Chinese Academy of Sciences (research at SIMIT, CAS) — Shanghai, China

M.S., Electrical and Computer Engineering — 2017 – 2019
Johns Hopkins University — Baltimore, MD, USA

B.S., Electrical and Computer Engineering — 2012 – 2017
Shanghai Jiao Tong University — Shanghai, China

RESEARCH INTERESTS

Embodied AI; multimodal large language models; occlusion / partial-observability reasoning; sensor intelligence and robust perception; generative vision and video understanding; edge UAV systems.

SELECTED PUBLICATIONS

* Equal contribution. Full list: Google Scholar / chaoyiwang09.github.io/publications

Qingdong He*, **Chaoyi Wang***, Peng Tang, Yifan Yang, Xiaobin Hu. "CLEAR: Context-Aware Learning with End-to-End Mask-Free Inference for Adaptive Video Subtitle Removal." *International Conference on Machine Learning (ICML)*, 2026. **Oral & Spotlight**.

Qingdong He*, Xueqin Chen*, **Chaoyi Wang**, Yanjie Pan, Xiaobin Hu, Zhenye Gan, Chengjie Wang, Xiangtai Li, Jiangning Zhang, Yabiao Wang. "Reasoning to Edit: Hypothetical Instruction-Based Image Editing with Visual Reasoning." *International Conference on Machine Learning (ICML)*, 2026.

Chaoyi Wang, Qingdong He, Yifan Yang, Jun Pei, Lijie Xia, Jianpo Liu, Baoqing Li. "PERCH: Perception–Evidence–Reinforcement CHain for Open-Vocabulary Aerial Object-to-Viewpoint Navigation." *Submitted to AAAI 2027*, 2026.

Chaoyi Wang, Qingdong He, Yifan Yang, Jun Pei, Lijie Xia, Jianpo Liu, Baoqing Li. "STARE: Spend To Acquire Reliable Evidence for Risk-Controlled Target Selection and Navigation on Edge UAVs." *Submitted to AAAI 2027*, 2026.

Chaoyi Wang, Fangzhou Meng, Jun Pei, Lijie Xia, Jianpo Liu, Xiaobing Yuan, Xinhan Di. "OCC-MLLM-CoT: Self-correction enhanced occlusion recognition with large language models via 3D-aware supervision and chain-of-thoughts guidance." *Image and Vision Computing*, 167:105881, 2026.

Chaoyi Wang, Qingdong He, Jun Pei, Lijie Xia, Jianpo Liu, Baoqing Li, Xinhan Di. "OCC-MLLM-V1: Occlusion reasoning with commonsense-guided Multi-modal LLM based agent via internal Chain-of-Thoughts (CoTs)." *Computer Vision and Image Understanding*, 270:104828, 2026.

Chaoyi Wang, Qingdong He, Yifan Yang, Jun Pei, Lijie Xia, Jianpo Liu, Baoqing Li, Xinhan Di. "OCC-MLLM-V2: Joint understanding and generation for occluded objects via multi-modal token learning." *Journal of Visual Communication and Image Representation*, 119:104887, 2026.

Chaoyi Wang, Yaozhe Song, Hongying Tang, et al. "Real-time vehicle sound detection system based on depthwise separable convolution neural network and spectrogram augmentation." *Remote Sensing*, 14(19):4848, 2022.

EXPERIENCE

Sensor Algorithm Engineer, Research Department — Jul 2020 – Present
Shanghai Institute of Microsystem and Information Technology / UCAS — Shanghai, China

- Develop multimodal signal-processing and target-detection algorithms for acoustic, vibration, image, and infrared sensing under real-world noise and resource constraints.

- Lead first-author embodied UAV research (PERCH / STARE, submitted to AAAI 2027): viewpoint-conditioned aerial ObjectNav and spend-to-verify risk-controlled commitment, including edge multirotor field tests.
- Prototype algorithms in Python/MATLAB and migrate high-performance modules to embedded platforms for field systems.

Speech Algorithm Engineer, AI Education — May 2019 – Jul 2020
Shanghai Leyan Technology Co., Ltd. — Shanghai, China

- Built Mandarin/English ASR and pronunciation-assessment (GOP) systems for K–12 English learning products.

SELECTED PATENTS & AWARDS

- Patents (selected): long-video generation with background transition (2023); field moving-vehicle detection with CRNNs (2023); robust moving-target counting with sensor arrays (2021).
- NVIDIA–Alibaba Cloud Heterogeneous Computing TensorRT AI Inference Hackathon Excellence Award, 2021.

SKILLS

Research: multimodal LLMs, diffusion / video generation, occlusion reasoning, robust sensing, computer vision.
Engineering: Python, PyTorch, C/C++, MATLAB; dataset construction; edge-oriented algorithm deployment.